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Re: Chaurasia et al: Highly Accurate and Precise Automated Cup-to-Disc Ratio Quantification for Glaucoma Screening

TO THE EDITOR: We read with interest the paper by Chaurasia et al[1] published in the September issue of *Ophthalmology Science*. The authors sought to validate a model to determine image gradability and estimate cup-to-disc ratio (CDR). They reported an impressive Pearson *R* of 0.93, indicating a strong correlation between predicted and expert-graded CDR. It is also worth noting that the authors made the effort to externally validate the model on different geographical populations by using the Drishti dataset from India.

To further improve the robustness and reproducibility of the CDR measurement, we suggest that the authors consider and evaluate how the model performs in the presence of peripapillary lesions, such as peripapillary atrophy or colobomas. In our study published in *Translational Vision Science and Technology* in April 2024,[2] we evaluated the performance between groups with and without peripapillary atrophy and found that the difference was statistically insignificant. The presence of other potential confounders such as cotton wool spots and nerve fiber layer reflection also did not seem to affect performance. However, we noted that our model might underperform in delineating accurate disc boundaries when images have unusual abnormalities or are underexposed. It would be interesting to see how the authors' model performs in the presence of these lesions, which would be particularly applicable in East Asian countries where myopia and its associated peripapillary atrophy are highly prevalent.

We also suggest potential changes to circumvent limitations in model performance. Firstly, we noted that the authors drop the poor-performing images during each epoch. Instead of dropping the images, we suggest weighting the loss terms for these poor-performing images.[3] This will enable the model to pay more attention to these images, reducing imbalanced classes and improving result reliability and performance.

Secondly, it is not explicitly stated in the paper which batch size and optimizer were used for model selection. We propose that each model be trained with multiple batch sizes and optimizers, as the convergence of different models might vary depending on the choice of batch size and optimizer.[4] This ensures that each model architecture is trained with the optimal optimizer and batch size.

Although the model holds significant promise in CDR estimation, we would be interested in the generalizability of its performance across different demographics and geographical locations, especially with varying optic disc morphologies. We also hope that with modifications to the methodology, the authors' model performance can be further enhanced, potentially paving the way for its improved utility in glaucoma screening.

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Disclosure(s):
All authors have completed and submitted the ICMJE disclosures form. The author(s) have no proprietary or commercial interest in any materials discussed in this article.

Available online: April 9, 2025.

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